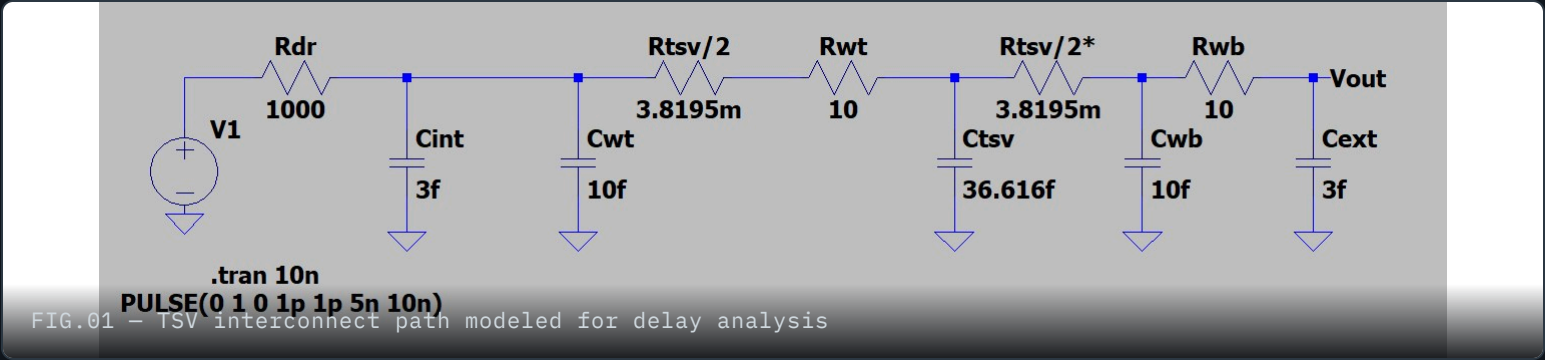


# Interconnect Delay & TSV Reliability

Two focused modeling exercises: SPICE-based interconnect delay analysis, and COMSOL-based thermomechanical stress simulation of through-silicon vias (TSVs) — two of the building blocks behind the packaging reliability work in the rest of this portfolio.



-45%

DELAY DROP: LUMPED RC VS. 10-SEGMENT MODEL

+41%

TSV PATH DELAY WHEN LINER K DOUBLES (2→4)

400°C

THERMAL CYCLING RANGE SIMULATED FOR TSVS

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LINER MATERIALS COMPARED FOR RELIABILITY

## ROLE

Independent coursework — both exercises modeled, run, and analyzed solo.

## CONTEXT

Advanced Packaging & Interconnect Modeling — Aalto University, Sep–Oct 2024.

## TOOLS

- SPICE
- COMSOL Multiphysics
- Elmore Delay Model

## Why a lumped model lies to you

The simplest way to estimate an interconnect's delay is to lump its total resistance and capacitance into one R and one C. It's also wrong — real wires are distributed, and modeling them as a chain of smaller RC

segments (or a single equivalent pi-network) gets meaningfully closer to what SPICE simulation actually shows.

LUMPED RC

Single lumped R-C model

Simplest model — overestimates delay



L = 1000  $\mu$ m

10.0 ps

5-SEGMENT

Distributed RC, N=5

Elmore delay, 5 segments



L = 1000  $\mu$ m

6.0 ps

10-SEGMENT

Distributed RC, N=10

Converging toward the true distributed delay



L = 1000  $\mu$ m

5.5 ps

$\pi$ -EQUIVALENT

Pi-equivalent model

Matches the 10-segment distributed result closely



L = 1000  $\mu$ m

5.49 ps

■ Propagation delay for a 1000 $\mu$ m line (10 $\Omega$ , 1pF total), scale relative to the lumped model

## Extending to a TSV interconnect path

The second half of the exercise applied the same delay analysis to a full **TSV interconnect path** — driver, wire, via, and receiver in series — using a TSV geometry of  $L = 30\mu\text{m}$ ,  $D = 10\mu\text{m}$ ,  $t = 1\mu\text{m}$  ( $R = 7.64\text{ m}\Omega$ ) and comparing two liner dielectric constants,  $k = 2$  and  $k = 4$ .

The analytical Elmore-delay calculation and the SPICE simulation agreed closely: **30.95 ps analytical vs. ~32 ps simulated** for  $k = 2$ , and **43.7 ps vs. ~44 ps** for  $k = 4$ . Doubling the dielectric constant roughly doubles the TSV's capacitance but only increases total path delay by about 41%, since the driver and wire resistances dominate the rest of the path.

$$t_p = 0.69 \cdot R_{dr} C_{ext} + 0.69 \cdot (R_{dr} + R_{wb}) C_{wb} + 0.69 \cdot (R_{dr} + R_{wb} + 0.5 R_{tsv}) C_{tsv} + 0.69 \cdot (R_{dr} + R_{wb} + R_{tsv} + R_{wt}) (C_{wt} + C_{int})$$



FIG.02 – SPICE delay simulation,  $k = 2$  ( $\approx 32\text{ ps}$ , matching the  $30.95\text{ ps}$  analytical result)

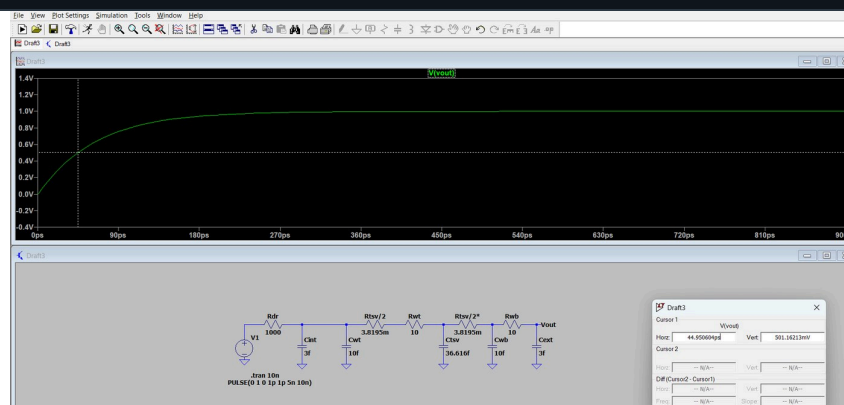


FIG.03 – SPICE delay simulation,  $k = 4$  ( $\approx 44\text{ ps}$ , matching the  $43.7\text{ ps}$  analytical result)

The takeaway carries directly into packaging material choice: a lower- $k$  dielectric around the TSV keeps propagation delay down and preserves signal integrity, at whatever trade-off that material brings elsewhere (see the liner discussion below).

## The via that stresses everything around it

Through-silicon vias give 3D-integrated chips their vertical connections — shortening interconnects, boosting bandwidth, and improving thermal paths. They also inject copper, a very different material to silicon, straight through the die, and that mismatch has consequences under thermal cycling. The second exercise used COMSOL FEA to simulate stress and displacement around single and multi-via TSV structures across a  $25^{\circ}\text{C} \rightarrow 400^{\circ}\text{C} \rightarrow 25^{\circ}\text{C}$  thermal cycle.

### Why TSVs need a liner

A dielectric liner electrically isolates the copper via from the surrounding silicon, blocks copper diffusion into the substrate, and provides mechanical stability. It needs high thermal stability, strong adhesion to both silicon and metal, and good insulating performance —  $\text{SiO}_2$  is the conventional choice.

### The keep-out zone

design cost

Mechanical stress from TSV formation and thermal cycling degrades nearby transistors, so no active circuitry can be placed within a **keep-out zone (KOZ)** around each via. In dense 3D-IC layouts with many TSVs, that KOZ eats directly into usable die area — a real trade-off between interconnect density and transistor density.

### Where the stress concentrates

Simulation showed the highest stress consistently at the **Cu– $\text{SiO}_2$  interface**, driven by the mismatch in thermal expansion between copper and oxide — copper expands more, putting itself in tension and the oxide in compression. Displacement peaks at the top of the via and tapers toward the base.

### Vias interact when packed close

In a 4-via array, the individual stress fields of neighboring TSVs overlap. Peak stress concentrates immediately around each via, while the region enclosed by the four vias stays comparatively low-stress — array layout, not just via geometry, shapes the reliability picture.

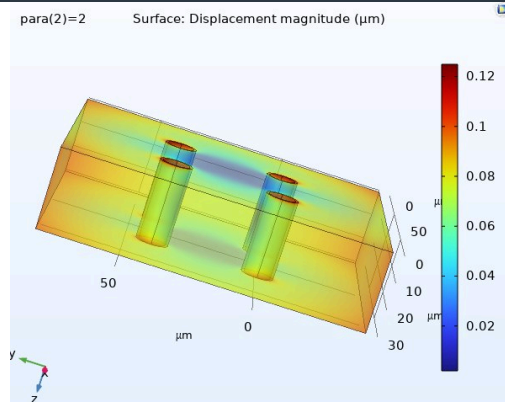


FIG.04 – Single-via COMSOL simulation: stress concentrates at the Cu-SiO<sub>2</sub> interface

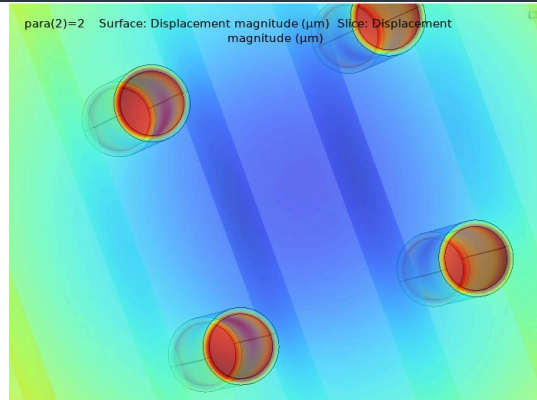


FIG.05 – Four-via array: overlapping stress fields around each TSV under 400°C cycling

### SiO<sub>2</sub> vs. a low-k liner

material trade-off

Swapping in a low-k liner — better CTE-matched to copper — lowers interfacial stress at the Cu-liner boundary, but its own higher thermal expansion produces **larger displacement** and a correspondingly **larger keep-out zone**. Lower stress at the interface isn't free: it trades against more physical movement and more silicon given up around every via. The right choice depends on whether a design is stress-limited or area-limited.

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ENGINEERING DECISION

## There's no single "right" TSV liner — it depends what you're optimizing for

The simulation didn't converge on one liner material as universally best. It surfaced a genuine trade-off between two failure modes, and which one dominates depends on the rest of the design.

#### OPTION A — SiO<sub>2</sub> (CONVENTIONAL)

The standard liner choice: high thermal stability, strong adhesion to both silicon and copper, good insulating performance. Larger CTE mismatch with copper concentrates stress at the Cu–SiO<sub>2</sub> interface.

#### OPTION B — LOW-K LINER

Better CTE-matched to copper, which lowers interfacial stress at the Cu–liner boundary. But its own higher thermal expansion produces larger displacement under cycling — and a correspondingly larger keep-out zone around every via.

#### WHY THERE'S NO SINGLE ANSWER

Lower interfacial stress isn't free — it trades against more physical movement and more silicon given up around every via. A stress-limited design (worried about interface cracking) favors the low-k liner; an area-limited design (dense 3D-IC layout where keep-out zones eat directly into usable die area) favors conventional SiO<sub>2</sub>. The right choice is a function of the rest of the chip, not the TSV in isolation.

## 04 CHALLENGES

### Where the modeling got genuinely tricky

#### Reconciling SPICE against the closed-form Elmore delay model

The analytical Elmore-delay calculation and the SPICE simulation agreed closely but not exactly — 30.95 ps analytical vs. ~32 ps simulated at k=2, 43.7 ps vs. ~44 ps at k=4. Elmore delay is a first-order approximation that collapses a distributed RC line into a simplified moment calculation; SPICE captures the full distributed response. Understanding *why* the small residual gap exists, rather than treating it as simulation noise, mattered more than the numbers matching.

#### Making sure the interface stress concentration was physical, not a meshing artifact

The COMSOL model consistently shows peak stress at the Cu–SiO<sub>2</sub> interface, driven by the CTE mismatch under thermal cycling. Stress concentrations at material interfaces are exactly where a coarse or poorly-refined mesh can produce numerically inflated results, so confirming the peak was a real physical effect — not a meshing artifact — required deliberate mesh refinement at that boundary rather than trusting the first result.

## 05 REFLECTION

### What I'd do with another pass at this

### Next steps

The liner trade-off was only characterized for a single via and a 4-via array. A natural next step would be running the low-k liner option through the same 4-via array simulation to see whether the stress-vs-keep-out-zone trade-off holds at higher via density, since the case study already shows that neighboring TSVs' stress fields overlap — the interaction itself might shift which liner is preferable as density increases, not just the absolute numbers.

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## SUMMARY

### What these exercises demonstrate

- SPICE circuit simulation and validation against analytical (Elmore) delay models
- COMSOL multiphysics FEA: thermal-structural coupling under cyclic loading
- Connecting circuit-level and mechanical-level results back to packaging design decisions
- Distributed vs. lumped RC modeling and when each approximation breaks down
- Reasoning about materials trade-offs (dielectric constant, CTE mismatch)
- Clear technical documentation of quantitative simulation results